# Subhadeep Pal

Postdoctoral Associate · Massachusetts Institute of Technology

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## Academic appointments

**Massachusetts Institute of Technology · Jan 2026-present**

Postdoctoral Associate, Civil & Environmental Engineering
Laboratory for Atomistic and Molecular Mechanics · Advisor: Markus Buehler

## Education

**Northwestern University · 2020-2025**

PhD in Mechanics, Materials, and Structures, Civil & Environmental Engineering
Advisor: Sinan Keten

**Indian Institute of Technology Kharagpur · 2015-2020**

Integrated BTech and MTech in Civil & Structural Engineering

## Research interests

Multiscale fracture mechanics; polymer nanocomposites and networks; molecular simulation; active learning and inverse design; traceable scientific AI.

## Honors and research support

2025 · Padden Award Symposium Finalist, APS Division of Polymer Physics

2023-2025 · Ryan Fellowship, International Institute of Nanotechnology, Northwestern

2023-2025 · NSF ACCESS computational allocation: 400,000 compute hours; proposal author and Computational Investigator

2025 · NSF Student Travel Award, Society of Engineering Science

2025 · Student Travel Award, USNCCM18

2024 · Graduate School Spotlight, Northwestern

2020-2021 · Murphy Fellowship, Northwestern

2019 · UTSIP Scholarship, University of Tokyo

# Subhadeep Pal

## Published journal articles · 2024-2026

### Molecular Investigation of Freshly-made Cement Paste Rheology for 3D-printing Application

**Pal, S.**; Gomaa, S.; Cusatis, G.; Keten, S.

*Journal of Applied Physics · 2026 · Editor's Pick*

### Process-Driven Control of the Properties of Polymer Grafted Nanoparticle Composites

Singh, M.; Lin, H.; **Pal, S.**; Zhou, C.; Mohammed, M.; Maddukuri, S.; Deen, P.; Yin, R.; Bockstaller, M.; Matyjaszewski, K.; Smith, Z.; Karim, A.; Kysar, J.; Keten, S.; Benicewicz, B.; Kumar, S. K.

*ACS Nano · 2026*

### From Packing to Performance: Decoding Nanoscale Mechanics of Binary Polymer-Grafted Nanoparticles via Active Learning

**Pal, S.**; Khumbhojkar, G.; Chen, W.; Keten, S.

*Macromolecules · 2026*

### Nanoconfinement Release Toughens Polymer Grafted Nanoparticles Assemblies Through Better Interdigitation and Entanglements

Moussavi, A.; Wu, Z.; **Pal, S.**; Keten, S.

*Nano Letters · 2025*

### Real-Time Visualization of Polymer Single Chain Conformational Change in Bulk State Using Super-Resolution Microscopy

Xiao, R.; **Pal, S.**; Rademacher, C.; Chen, J.; Wang, Q.; Chen, W.; Keten, S.; Shull, K. R.; Wang, M.

*Physical Review Letters · 2025 · Editor's Suggestion*

### Engineer Reversible Nanocomposite by Programming Amorphous Polymer Conformation Under Nanoconfinement

Chen, T.; Qian, Y.; Laine, A.; Kwon, J.; Liu, L.; **Pal, S.**; Gupta, S.; Vargo, E.; Su, G. M.; Ritchie, R. O.; Keten, S.; Wang, R.; Salmeron, M.; Xu, T.

*Advanced Materials · 2025*

### Network Topology and Percolation in Model Covalent Adaptable Networks

Hafner, B.; **Pal, S.**; Lewis, B.; Keten, S.; Shull, K.

*ACS Macro Letters · 2024 · Journal cover*

### Micro-Ballistic Response of Thin Film Polymer Grafted Nanoparticle Monolayers

**Pal, S.**; Keten, S.

*Soft Matter · 2024 · Pioneering Investigators series*

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## Published journal articles · continued

### Characterizing the Shear Response of Polymer-Grafted Nanoparticles

Moussavi, A.; **Pal, S.**; Wu, Z.; Keten, S.

*Journal of Chemical Physics · 2024 · Polymer Nanoconfinement special collection*

### Predictive Molecular Modeling of Hardener Composition Effects on Epoxy Resin Fracture Properties

**Pal, S.**; Dansuk, K. C.; Giuntoli, A.; Sirk, T.; Keten, S.

*Macromolecules · 2023*

### A Catch Bond Mechanism with Looped Adhesive Tethers for Self-Strengthening Materials

Dansuk, K. C.; **Pal, S.**; Keten, S.

*Communications Materials · 2023*

### Implicit Chain Particle Model for Polymer-Grafted Nanoparticles

Wu, Z.; **Pal, S.**; Keten, S.

*Macromolecules · 2023*

### Shock Wave Propagation Through Air: A Reactive Molecular Dynamics Study

**Pal, S.**; Mitra, N.

*Proceedings of the Royal Society A · 2021*

### Stretch-Induced Helix to Extended Coil Transition of Crystalline alpha Phase Isotactic Polypropylene: A Molecular Dynamics Study

**Pal, S.**; Mitra, N.; Sarkar, P. K.; Prasad, D.

*Polymer Crystallization · 2020 · Journal cover*

## Accepted / in press

### Graph-Native Reinforcement Learning Enables Traceable Scientific Hypothesis Generation through Conceptual Recombination

**Pal, S.**; Sourav, S.; Ghosal, T.; Buehler, M. J.

*AI and Science · 2026 · In press*

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## Preprints and manuscripts under review

### SwarmWorld: Stigmergic Technological Evolution in Societies of Language-model Agents

**Pal, S.**; Wang, F. Y.; Buehler, M. J.

*arXiv · 2026 · Under review*

### Hierarchical Multi-Agent Reasoning Systems for Knowledge-Grounded Material Substitution

Hage, T. P.*; **Pal, S.***; Lu, W.; Hsu, Y.-C.; Lyon, G.; Jin, J.; Buehler, M. J.

*arXiv · 2026 · Equal contribution · Under review*

### Autonomous Agents Coordinating Distributed Discovery Through Emergent Artifact Exchange

Wang, F. Y.; Marom, L.; **Pal, S.**; Luu, R. K.; Lu, W.; Berkovich, J. A.; Buehler, M. J.

*arXiv · 2026 · Under review*

## Conference and workshop contributions

### Visualizing Graph-to-Answer Mechanism Recovery in Materials-Science Hypothesis Generation

Sourav, S.; **Pal, S.**; Buehler, M. J.; Das, S.; Wang, F. Y.; Soos, D.; Ghosal, T.

*IEEE VIS 2026 Workshop · 2026 · Preprint*

### Graph-to-Answer Circuits: Tracing Scientific Reasoning Inside Graph-Native AI Co-Scientists in Material Science

Sourav, S.; **Pal, S.**; Buehler, M. J.; Ghosal, T.

*ACM AI Summit · 2026*

### Integration of Image Processing and Crowd Simulation for Understanding Crowd Dynamics in Kumbh Mela

Karthika, P. S.; Verma, A.; **Pal, S.**

*Transportation Research Procedia · 2020*

Publication status follows the October 2026 faculty-application CV. * denotes equal contribution.

# Subhadeep Pal

## Teaching, mentoring and service

**MIT · Atomistic Modeling and Simulation · Fall 2026**

Graduate Teaching Assistant; taught sessions, held office hours, evaluated problem sets, and advised 30 students.

**Northwestern · Finite Element Methods in Mechanics · Fall 2022**

Graduate Teaching Assistant; coordinated TA work, evaluated assignments, and supported 36 students during office hours.

**Northwestern · Structural Analysis - Dynamics · Fall 2021**

Grader; assessed assignments and examinations and advised students on course projects.

## Research mentoring

MIT: graduate and undergraduate mentoring (2026-present), alongside high-school research mentoring through the Research Science Institute. Northwestern: doctoral mentoring in civil and mechanical engineering across 2022-2025.

## Leadership

**ScienceClaw Hackathon · MIT · 2026**

Co-organized a hackathon with 100 academic and industry participants; helped secure $200,000 in sponsorship. Supported by the E14 Seed Fund and industry partners.

**Graduate International Student Association · Northwestern · 2021-2025**

Treasurer and event organizer; managed budgets and secured $16,000 in funding from the Office of Student Life.

**Bengalis in Chicago · UIC · 2021-2024**

Event organizer; coordinated volunteers and logistics for cultural events with more than 500 attendees.

## Peer-review service

Nature Chemistry; International Journal of Impact Engineering; Journal of Chemical Physics; Soft Matter; Nanoscale Advances; Journal of Physical Chemistry; npj Computational Materials; Journal of the Mechanical Behavior of Biomedical Materials; Journal of Chemical Information and Modeling.

Conference presentations and detailed research summaries: palsubhadeep1234.github.io/home/cv/